

yolov10n Final Training Report

System Information

OS	Linux 6.6.105+
Python Version	3.12.12
PyTorch Version	2.9.0+cu126
CUDA Available	True
Device	cuda
RAM (GB)	13.61

Dataset Information

Property	Value
Dataset	bdd100k_yolo_limited
Number of Classes	10
Train Images	945
Val Images	611
Test Images	725
Data YAML	data.yaml

Classes:

pedestrian, rider, car, truck, bus, train, motorcycle, bicycle, traffic light, traffic sign

Optimization Summary

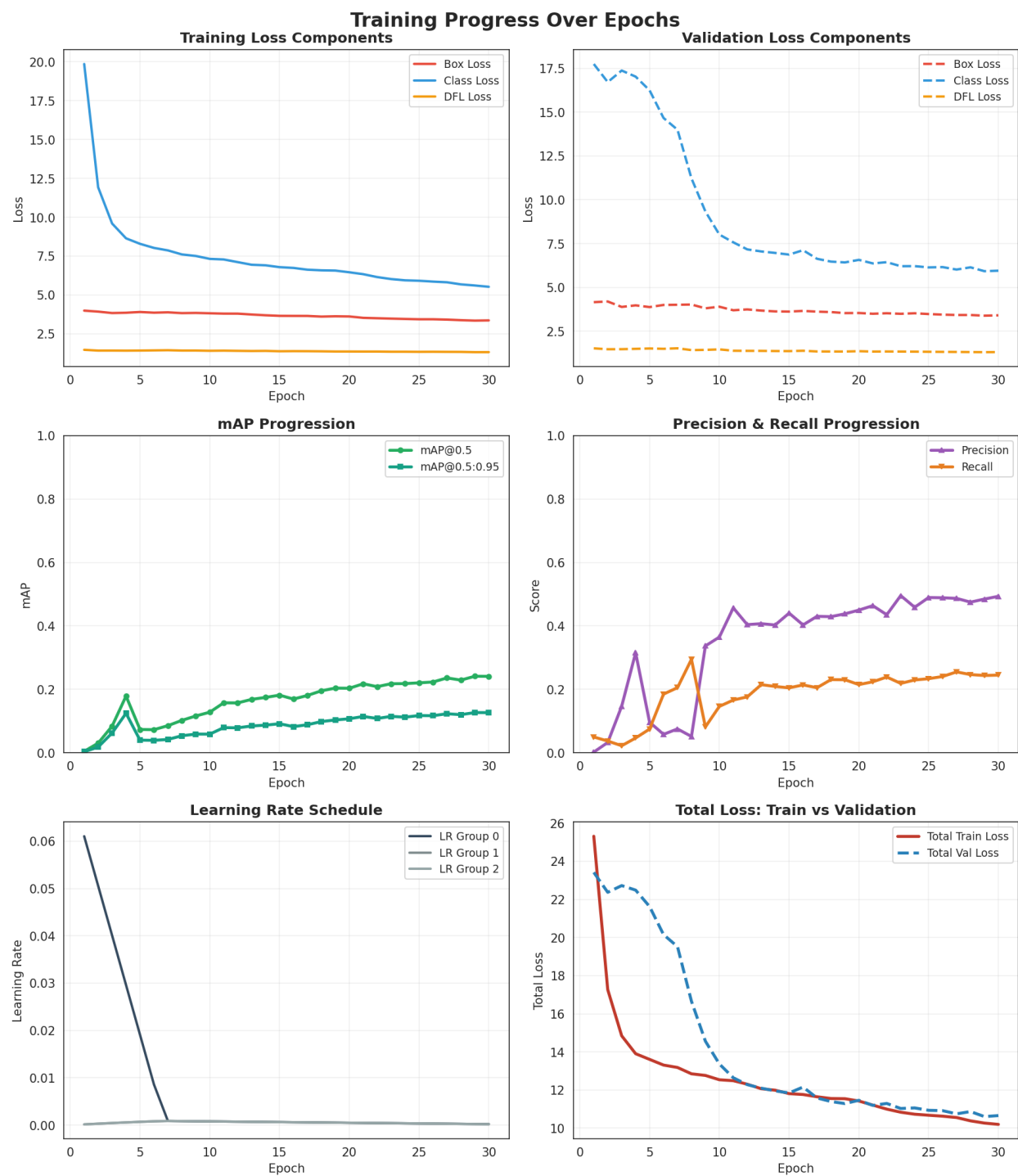
Metric	Value
Total Trials	12
Best Trial Number	0
Best Trial mAP@0.5	0.1530
Epochs per Trial	6
Final Training Epochs	30

Optimized Hyperparameters Used

Parameter	Value
optimizer	Adam
lr0	0.001040
lrf	0.162407
momentum	0.868719
weight_decay	0.000013
warmup_epochs	3
warmup_momentum	0.770502
warmup_bias_lr	0.070807
hsv_h	0.002058
hsv_s	0.787964
hsv_v	0.732977
scale	0.827403
translate	0.036365
mosaic	0.591702
mixup	0.060848
box	7.148539
cls	1.147918
df1	0.936844

Training Process Analysis

Epoch-by-Epoch Training Metrics



Detailed Epoch Metrics

Epoch	Train Loss	Val Loss	mAP@0.5	mAP@0.5:0.95	Precision	Recall
1	25.3147	23.4177	0.0042	0.0029	0.0018	0.0495
2	17.2664	22.3737	0.0301	0.0177	0.0322	0.0367
3	14.8451	22.7261	0.0837	0.0620	0.1465	0.0219
4	13.9049	22.4881	0.1785	0.1243	0.3154	0.0463
5	13.6069	21.6263	0.0732	0.0396	0.0961	0.0745
6	13.3091	20.1534	0.0726	0.0389	0.0579	0.1848
7	13.1824	19.5226	0.0852	0.0417	0.0749	0.2051
8	12.8502	16.6593	0.1019	0.0534	0.0512	0.2944
9	12.7652	14.5626	0.1158	0.0587	0.3367	0.0813
10	12.5380	13.3670	0.1277	0.0584	0.3647	0.1459
14	11.9902	11.9547	0.1743	0.0869	0.4028	0.2089
15	11.8101	11.8453	0.1815	0.0913	0.4403	0.2045
16	11.7640	12.1563	0.1691	0.0822	0.4031	0.2137
17	11.6460	11.5827	0.1805	0.0882	0.4302	0.2050
18	11.5544	11.3956	0.1953	0.0981	0.4292	0.2305
21	11.2080	11.1928	0.2173	0.1143	0.4638	0.2236
22	10.9975	11.2957	0.2083	0.1079	0.4352	0.2387
23	10.8352	11.0308	0.2175	0.1145	0.4955	0.2186
24	10.7307	11.0609	0.2179	0.1120	0.4587	0.2293
25	10.6768	10.9351	0.2203	0.1172	0.4895	0.2334
26	10.6269	10.9174	0.2230	0.1166	0.4888	0.2404
27	10.5566	10.7493	0.2358	0.1229	0.4868	0.2548
28	10.3828	10.8700	0.2290	0.1199	0.4750	0.2464
29	10.2650	10.6063	0.2411	0.1266	0.4846	0.2435
30	10.1968	10.6587	0.2409	0.1263	0.4934	0.2448

Training Statistics Summary

Metric	Initial	Final	Best	Change
mAP@0.5	0.0042	0.2409	0.2411	+0.2367
mAP@0.5:0.95	0.0029	0.1263	0.1266	+0.1234
Precision	0.0018	0.4934	0.4955	+0.4915
Recall	0.0495	0.2448	0.2944	+0.1953

Final Model Performance

Metric	Value
mAP@0.5	0.2417
mAP@0.5:0.95	0.1267
Precision	0.4864
Recall	0.2452

Test Set Validation Results

Model Architecture & Performance

Metric	Value
Model Name	yolov10n_finetuned_20251124
Parameters (M)	2.27
Model Size (MB)	4.48
FLOPs (G)	6.54
Layers	102
Inference Speed (FPS)	188.87
IoU Threshold	0.50

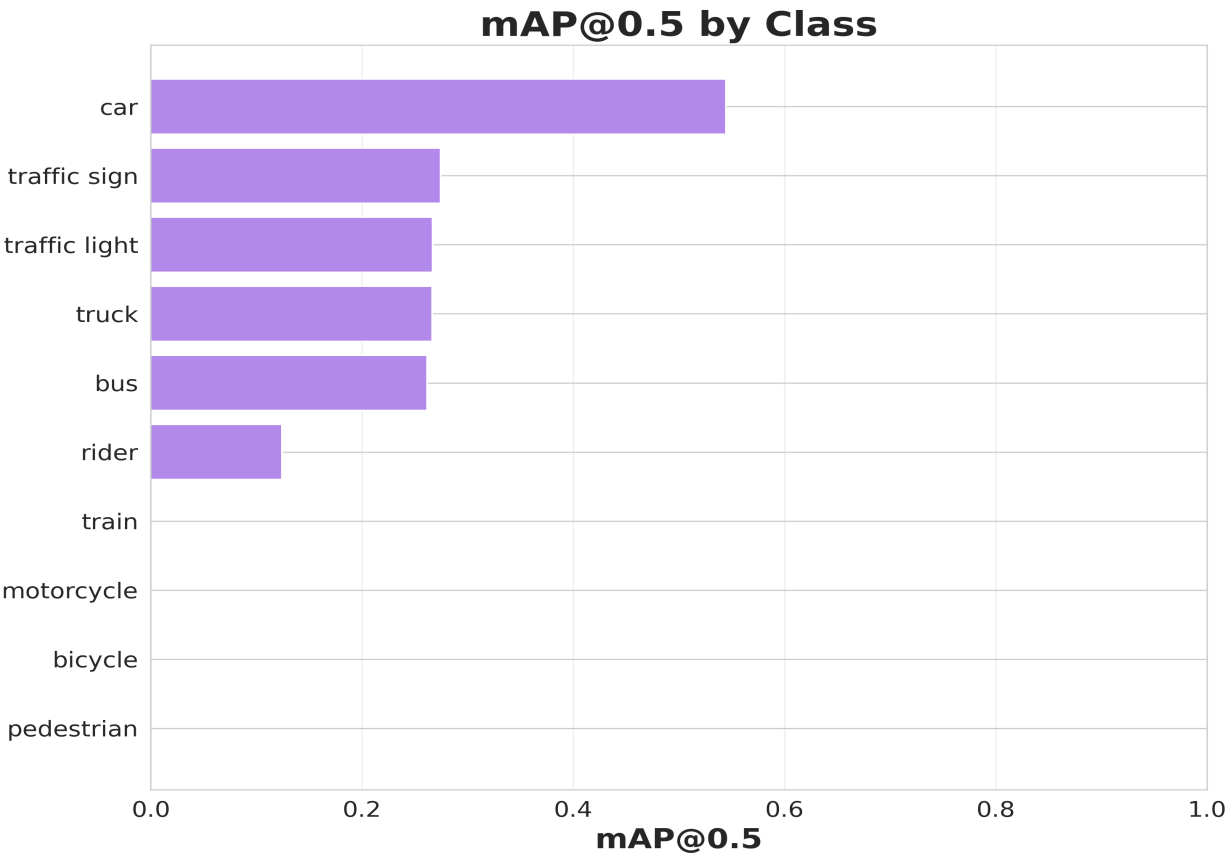
Overall Performance Metrics on Test Set

Metric	Confusion Matrix	YOLO Validation
Precision	0.3429	0.5278
Recall	0.7209	0.2555
F1-Score	0.4647	N/A
mAP@0.5 (Overall)	N/A	0.2484
mAP@0.5:0.95 (Overall)	N/A	0.1313

Per-Class Performance Metrics

Class	Precision	Recall	F1-Score	mAP@0.5	TP	FP	FN
pedestrian	0.0000	0.0000	0.0000	0.0000	0	0	0
rider	0.0281	0.3846	0.0524	0.1242	5	173	8
car	0.4578	0.7473	0.5678	0.5447	3986	4721	1348
truck	0.2011	0.5286	0.2913	0.2666	111	441	99
bus	0.1173	0.5195	0.1914	0.2619	40	301	37
train	0.0000	0.0000	0.0000	0.0000	0	17	0
motorcycle	0.0000	0.0000	0.0000	0.0000	0	0	0
bicycle	0.0000	0.0000	0.0000	0.0000	0	0	0
traffic light	0.1893	0.6957	0.2977	0.2671	487	2085	213
traffic sign	0.2017	0.6480	0.3077	0.2745	556	2200	302

mAP@0.5 Distribution by Class



Intersection over Union (IoU) Analysis

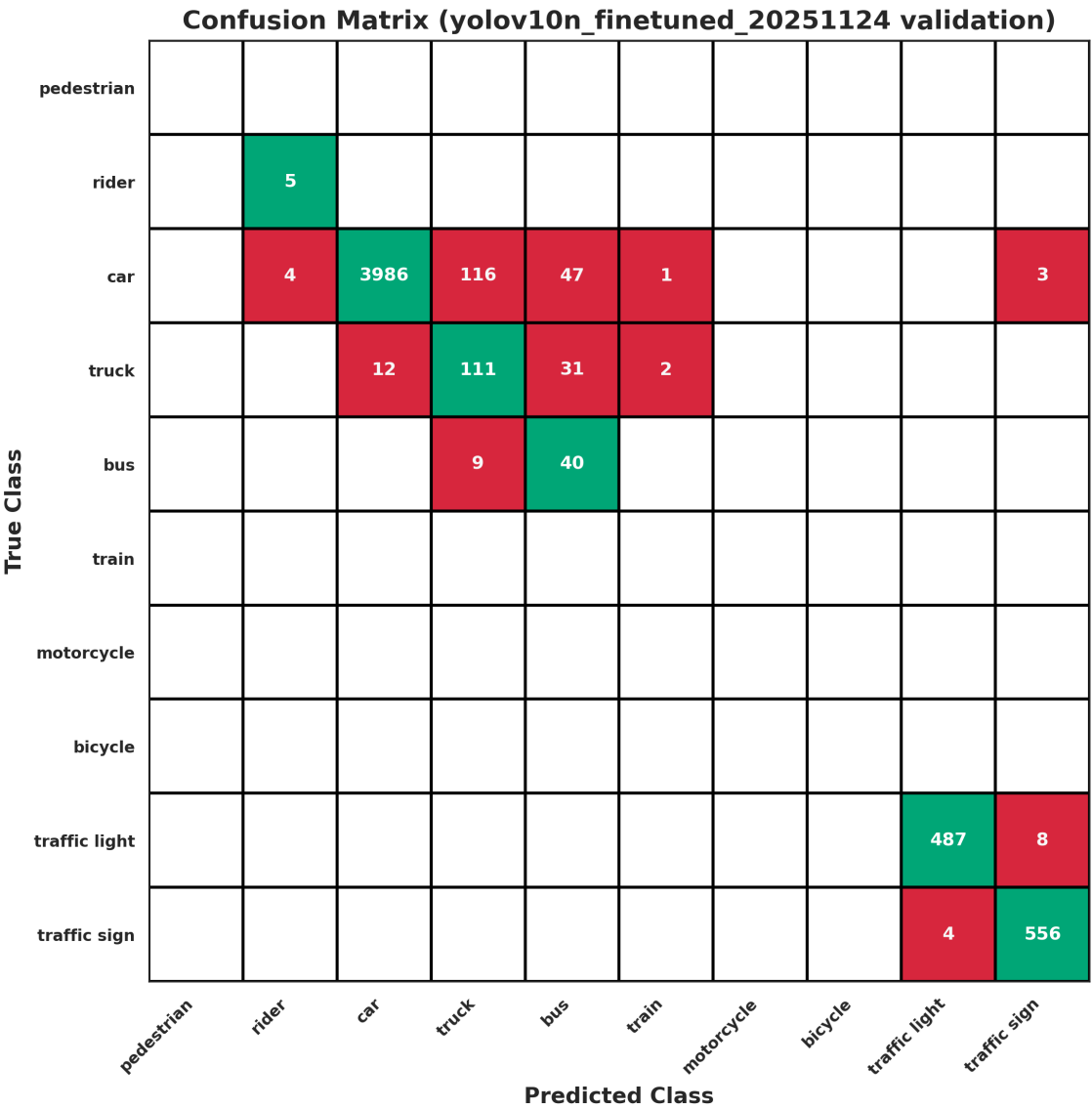
IoU Threshold Used: 0.50

IoU (Intersection over Union) measures the overlap between predicted and ground truth bounding boxes. A prediction is considered correct (True Positive) when $\text{IoU} \geq 0.50$.

Per-Class IoU Performance:

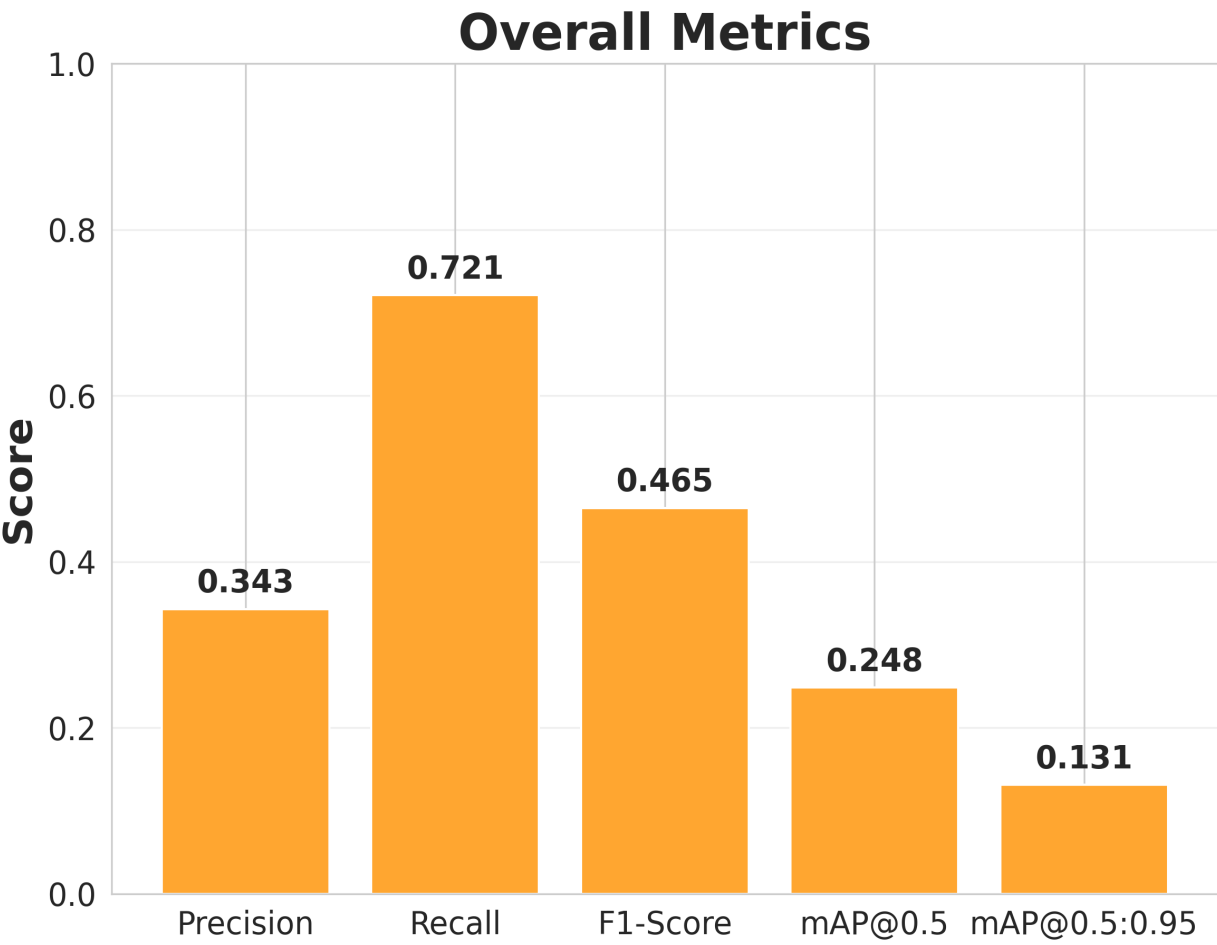
The confusion matrix and per-class metrics above show detection accuracy at IoU=0.50 threshold. Each class's True Positives (TP) represent detections with $\text{IoU} \geq 0.50$.

Confusion Matrix (Test Set)



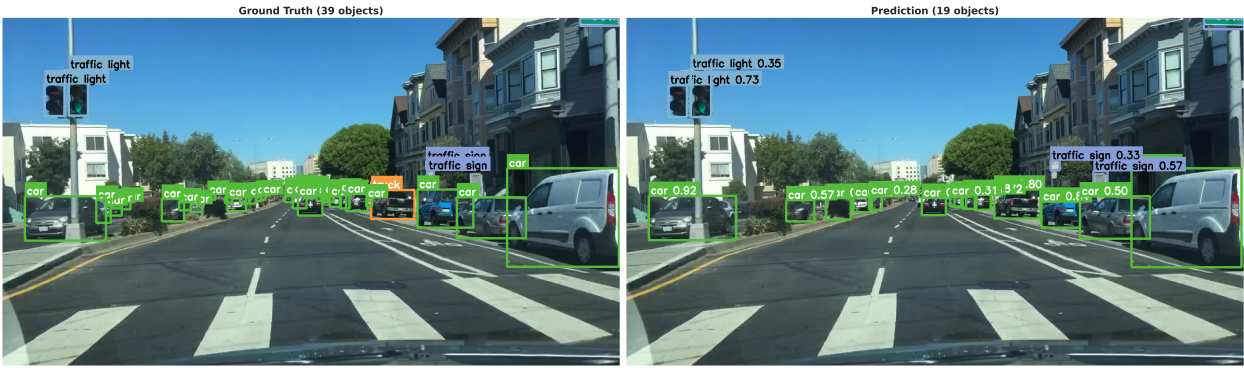
Test Set Performance Curves

Overall Metrics Visualization



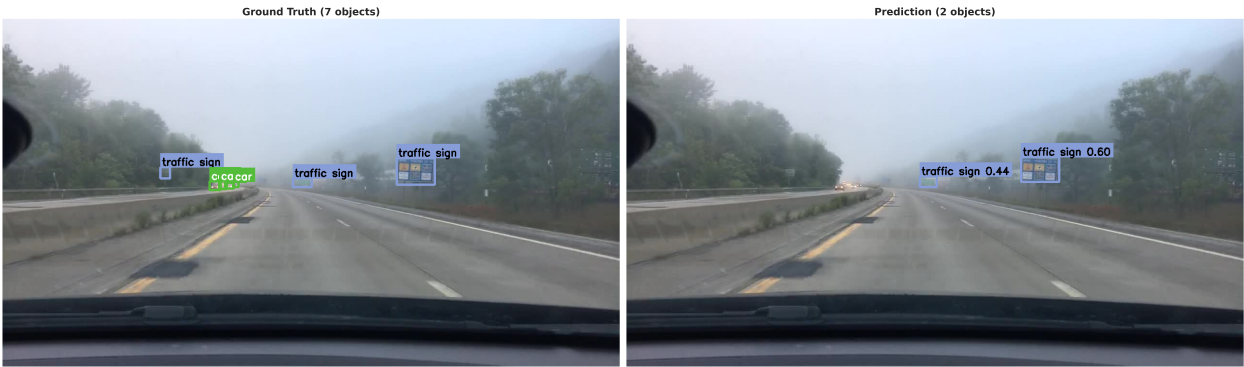
Sample Predictions: Ground Truth vs Model Output

Sample 1 - Weather: clear, Scene: residential, Time: daytime



Ground Truth: 39 objects | Predictions: 19 objects

Sample 2 - Weather: foggy, Scene: highway, Time: daytime



Ground Truth: 7 objects | Predictions: 2 objects

Sample 3 - Weather: undefined, Scene: undefined, Time: daytime



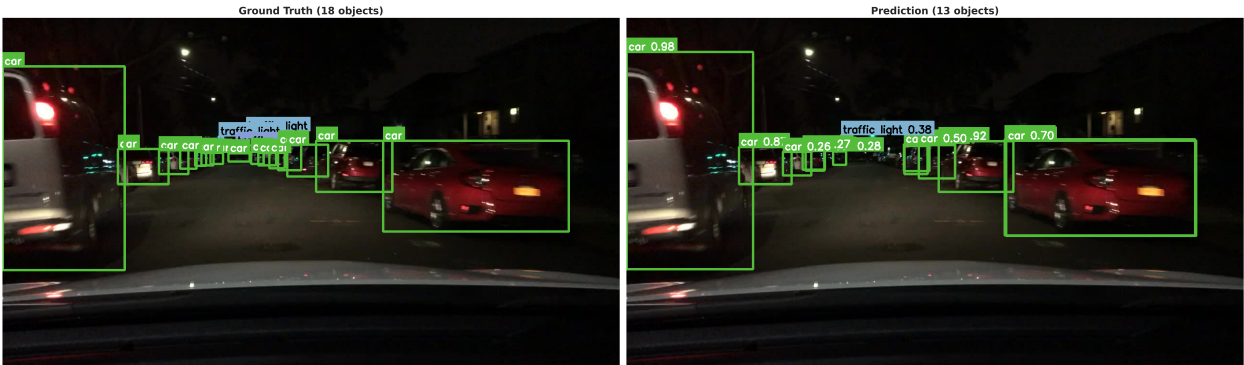
Ground Truth: 3 objects | Predictions: 0 objects

Sample 4 - Weather: rainy, Scene: highway, Time: daytime



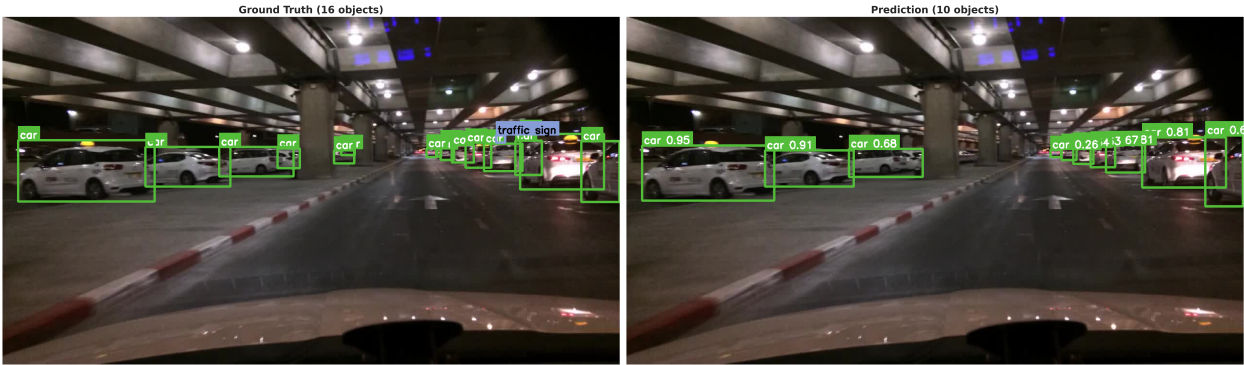
Ground Truth: 24 objects | Predictions: 13 objects

Sample 5 - Weather: clear, Scene: undefined, Time: night



Ground Truth: 18 objects | Predictions: 13 objects

Sample 6 - Weather: undefined, Scene: parking lot, Time: night



Ground Truth: 16 objects | Predictions: 10 objects